

PROJECT PLAN: Bid Management & Vendor Coordination System

MVP Scope

Prepared for: BluOnX / Capstone

Document Version: 2.0

Date: February 10, 2026

Document Owner: Awais Anwer (Tkrupt)

I. Overview & Summary

This document provides a comprehensive implementation plan for an MVP Bid Management and Vendor Coordination System for Blu OnX Development (land/horizontal) and Capstone LLC (vertical construction). The system streamlines the complete vendor lifecycle from bid invitation through contract award and milestone tracking, eliminating manual processes, ensuring standardized bid comparison, and preventing scheduling conflicts.

The implementation uses a custom secure form portal for bid submission with magic link authentication, providing complete control and a professional vendor experience. Post-award milestone tracking uses simple email-based check-ins to avoid overwhelming non-technical vendors. This Phase 1 delivery will reduce bid cycle time from weeks to days, eliminate manual spreadsheet juggling, and provide complete audit trails for all vendor interactions.

Key Objectives:

- Automate vendor bid invitation and tracking with intelligent filtering
 - Implement secure magic link authentication for bid submission
 - Standardize bid collection through custom secure portal
 - Enable side-by-side bid comparison with intelligent scoring
 - Implement pre-award validation to prevent scheduling conflicts
 - Provide real-time dashboard visibility into bidding status
 - Generate award letters and contracts automatically with DocuSign integration
 - Implement simple email-based milestone tracking with automated deadline alerts
 - Ensure production-ready quality through comprehensive testing
-

II. Tools & Technology Stack

Component	Technology	Purpose
Frontend	React + Vite + TailwindCSS	Admin dashboard & UI
Backend	FastAPI + Supabase DB	API & Database
Authentication	Supabase Auth	Email/password
Bid Collection	Custom React Portal + Magic Links	Vendor bid submission
Email Service	AWS SES or SendGrid Pro	Automated notifications & reminders
API Layer	React Query	Data fetching & caching
Testing	Playwright + Jest	Unit, integration, E2E testing
Automations	n8n Pro	Automation scheduling & workflows
Location Services	Google Maps API	Distance calculation, geocoding
E-Signatures	DocuSign API	Contract signing & execution

Hosting (Frontend)	Vercel Pro	Frontend deployment
Hosting (Backend)	AWS (EC2/ECS/Lambda)	Backend API hosting
Version Control	Git + GitHub	Code management

III. Real-Life Scenario

Imagine Samantha, a Project Manager at Blu OnX Development, starting a new excavation and grading project:

1. Project Creation

Samantha logs into the dashboard using her Google account (or email/password) and creates "Sunset Hills Phase 2" project with a \$2.5M budget, uploads site plans and specifications, and adds 8 tasks including excavation, utilities, and paving with individual budgets and required timelines.

2. Automated Bid Invitations

She clicks "Start Bidding" on the Excavation & Grading task (\$145K budget). The system automatically filters 7 qualified excavation vendors within 50 miles using Google Maps distance calculation who have current insurance, adequate bonding, and available capacity. It generates unique secure portal links with magic link authentication for each vendor with pre-filled company information, and sends professional bid invitation emails with all project documents attached.

3. Real-Time Tracking

Over the next week, the dashboard updates in real-time showing "3 of 7 vendors submitted." The system automatically sends friendly reminders at T-7 days, urgent reminders at T-3 days, and final call on the due date to non-responders via n8n automated workflows.

4. Intelligent Comparison

On the bid deadline, Samantha opens the comparison view. The system displays all 3 bids side-by-side with complete line-item breakdowns. It normalizes different pricing formats (ABC uses lump sum, Johnson uses unit pricing), calculates weighted scores based on: 40% price, 20% compliance (insurance/docs), 20% past performance, 10% capacity, 10% timeline alignment. The system recommends ABC Excavation with a 94.5 score vs 89.2 for Johnson and 87.8 for XYZ.

5. Pre-Award Validation

Samantha selects ABC Excavation and clicks "Award Contract." The system performs automatic validation: can start on required March 1st date, insurance certificate valid until December 2025, currently 3 of 5 active jobs (has capacity), bid amount of \$145K within 5% of budget, bonding capacity adequate. All checks pass with no warnings.

6. Automated Award with DocuSign

She confirms the award. Within seconds, the system: generates a detailed award letter with mobilization instructions and project milestones, sends congratulations email to ABC Excavation with DocuSign contract for e-signature, sends professional decline letters thanking the other 6 vendors for their submissions, updates ABC's capacity from 3/5 to 4/5 active jobs, creates the contract record with start/end dates, logs the entire transaction for audit trail.

7. Email-Based Milestone Tracking

ABC Excavation's project starts with three milestones: "Site Clearing Complete" (March 8), "Rough Grading Complete" (March 15), and "Final Grade & Compaction" (March 22).

5 days before March 8: ABC receives automated email: "Reminder: Site Clearing work starts in 5 days"

March 8 (start date): ABC receives email: "Did you start Site Clearing today? [Yes, started on time] [No, there's a delay]" - ABC clicks [Yes]

March 13 (5 days before end): ABC receives email: "Will you complete Site Clearing on time by March 8? [Yes, on track] [No, need more time]" - ABC clicks [No]

Immediately: Samantha receives alert email: "ABC Excavation reported delay on Site Clearing. Contact: John Smith 555-1234. Update schedule: [Click here]"

Samantha calls ABC: Weather delay, need 3 more days. She clicks the email link, updates end date to March 11.

March 11: ABC receives completion check: "Did you complete Site Clearing? [Yes, completed] [No, still working]" - ABC clicks [Yes]

Immediately: Samantha receives notification: "ABC Excavation completed Site Clearing on Sunset Hills Phase 2"

The same automated cycle repeats for "Rough Grading Complete" and "Final Grade & Compaction" milestones.

8. Result

What used to take Samantha 2-3 weeks of manual work now takes 10 minutes. She has complete confidence that ABC Excavation can actually start on March 1st, complete audit trail for compliance, standardized bids for true apples-to-apples comparison, and all vendors were treated professionally. The \$145K excavation work is awarded to the best-qualified vendor with legally binding e-signature.

Throughout the project, automated email check-ins keep ABC accountable without overwhelming them with complex portal logins. Proactive alerts ensure Samantha stays ahead of any delays, eliminating the constant email chasing and manual status updates that used to consume hours of her day. She moves on to the next task with full visibility and control.

IV. Development Phases

Phase 1: Foundation & Database Setup (60 Hours)

Module 1: Initial Setup and Database Design

Task 1.1: Initialize Supabase project and configure environment -4h

- Set up production and staging Supabase projects
- Configure environment variables and secrets management
- **Tools:** Supabase platform, environment setup

Task 1.2: Design and implement core database schema -25h

- Tables: vendors, projects, tasks, bid_packages, bid_invitations, bid_submissions, bid_line_items, contracts, document_storage, email_log, docusign_envelopes, milestones, milestone_responses, milestone_alerts, vendor_flags
- Relationships: foreign keys, junction tables, cascading rules
- **Tools:** PostgreSQL, Supabase database

Task 1.3: Create database indexes for query optimization -2h

- Index vendor trade, location, status fields
- Index project and task status, dates
- Composite indexes for common query patterns
- **Tools:** PostgreSQL

Task 1.4: Implement Row Level Security (RLS) policies -4h

- Company-level data isolation
- User role-based access control
- Vendor-specific data access rules
- **Tools:** Supabase RLS

Task 1.5: Set up database triggers -5h

- Auto-update timestamps on all tables
- Vendor capacity management (increment/decrement on award)
- Budget calculations and validation
- Bid submission status updates
- **Tools:** PostgreSQL functions, PL/pgSQL

Task 1.6: Configure Supabase Storage buckets -4h

- Project documents bucket (plans, specs, photos)
- Vendor documents bucket (insurance, licenses)
- Bid attachments bucket
- RLS policies for secure access
- **Tools:** Supabase Storage

Task 1.7: Set up authentication framework -5h

- Configure Supabase Auth providers (email)
- Session management and token refresh
- User metadata storage
- **Tools:** Supabase Auth configuration

Task 1.8: Create development and staging environments -4h

- Separate Supabase projects for dev/staging/prod
- Environment-specific configuration
- Database seeding scripts
- **Tools:** Supabase, environment config

Task 1.9: Configure AWS infrastructure -10h

- Set up AWS account and IAM roles
- Configure EC2/ECS or Lambda for FastAPI backend
- Set up load balancer and auto-scaling
- Configure CloudWatch monitoring
- **Tools:** AWS Console, AWS CLI

Acceptance Criteria

- Supabase project fully configured with production and staging environments
 - Complete database schema with 14+ tables, all relationships, and indexes
 - RLS policies protect sensitive data appropriately
 - Automated triggers handle capacity updates, timestamps, and budget calculations
 - Authentication framework configured for email
 - AWS infrastructure ready for backend deployment
 - Storage buckets configured with proper access controls
-

Phase 2: Frontend Foundation, Backend API & Authentication (100 Hours)

Module 2: React Application Setup, FastAPI Backend & Authentication

Task 2.1: Initialize React + Vite project with TailwindCSS -4h

- Project scaffolding with Vite
- TailwindCSS configuration and custom theme
- **Tools:** React, Vite, TailwindCSS

Task 2.2: Set up routing with React Router -6h

- Define route structure for all major pages
- Protected route components
- Navigation guards
- **Tools:** React Router

Task 2.3: Configure Supabase client for state management -5h

- Supabase client initialization
- Auth state management
- Real-time subscription setup
- **Tools:** Supabase JavaScript client

Task 2.4: Set up React Query for API layer and caching -6h

- React Query configuration
- Custom hooks for data fetching
- Cache invalidation strategies
- **Tools:** React Query, Axios

Task 2.5: Create reusable UI component library -15h

- Buttons, inputs, forms, selects, checkboxes
- Cards, modals, dialogs, alerts
- Tables with sorting and filtering
- Loading states and skeletons
- Navigation components
- Data visualization components
- Form validation components
- **Tools:** React, TailwindCSS

Task 2.6: Build FastAPI backend on AWS -25h

Detailed Breakdown:

- FastAPI project initialization and structure
 - Project folder structure, virtual environment, dependencies
 - Configuration management
- API endpoint scaffolding for all major resources
 - Vendors, projects, tasks, bids, contracts, DocuSign endpoints, milestones
 - RESTful route structure
- Supabase database connection
 - Connection pooling, environment variables
 - Database migration handling
- CORS configuration for frontend integration
 - Vercel domain whitelisting
 - Preflight request handling
- API authentication middleware
 - JWT validation from Supabase tokens
 - Protected route decorators

- Role-based authorization
- AWS deployment configuration
 - Dockerfile for containerization
 - ECS task definitions or Lambda configuration
 - Environment variable management

Tools: FastAPI, Python, Uvicorn, AWS SDK

Task 2.7: Build authentication system -25h

Detailed Breakdown:

- Email/password authentication flow
 - Login, register, email verification
 - Password reset flow
 - Session management
- Session management and persistence
 - Token storage strategy
 - Automatic token refresh
 - Logout and session cleanup
- Protected route guards and RBAC
 - Route protection HOCs
 - Role-based UI rendering
 - Permission checking utilities

Tools: Supabase Auth, React, FastAPI

Task 2.8: Create main dashboard layout with navigation -8h

- Sidebar navigation with icons
- Top header with user menu
- Breadcrumb navigation
- Responsive hamburger menu for mobile
- **Tools:** React, TailwindCSS

Task 2.9: Implement responsive design -6h

- Mobile (320px+), tablet (768px+), desktop (1024px+) breakpoints
- Touch-friendly UI elements
- Mobile navigation patterns
- **Tools:** TailwindCSS responsive utilities

Acceptance Criteria

- React application running locally and deployable to Vercel Pro

- FastAPI backend running on AWS infrastructure
 - Clean, responsive UI with consistent design system across all screen sizes
 - Email/password authentication fully functional with password reset
 - Google Sign-In working seamlessly with proper profile data capture
 - Session persistence working correctly (users stay logged in)
 - Protected routes redirect unauthenticated users to login
 - FastAPI backend properly integrated with Supabase database
 - CORS configured correctly for Vercel domain
 - API endpoints scaffolded and ready for business logic
 - Authentication working end-to-end with proper error handling
 - Routing structure in place for all major pages
 - Reusable component library ready for rapid development
-

Phase 3: Core Entity Management (60 Hours)

Module 3: Projects, Tasks, and Vendors CRUD

Tasks

Task 3.1: Build vendor management interface

- Vendor list view with search, sort, filter
- Add/edit vendor form with validation
- CSV import with field mapping and validation
- Vendor detail page with history
- Delete vendor with confirmation
- Advanced filtering and bulk operations
- **Tools:** React, FastAPI, Supabase API, CSV parsing library

Task 3.2: Implement project management

- Project list view with status indicators
- Create project form with document upload
- Edit project details and budget tracking
- Project detail page with task list
- Budget allocation and tracking
- Project timeline visualization
- **Tools:** React, FastAPI, Supabase API

Task 3.3: Create task management within projects

- Add task form within project context
- Edit and delete tasks
- Task dependencies and sequencing
- Budget allocation per task
- Timeline and deadline setting
- **Tools:** React, FastAPI, Supabase API

Task 3.4: Build document upload functionality

- Drag-and-drop file upload component
- File type validation (PDF, JPEG, PNG, DWG)
- File size limits and compression
- Document preview and download
- Secure URL generation
- **Tools:** React, Supabase Storage, file handling libraries

Task 3.5: Integrate Google Maps for location

- Address geocoding for vendors and projects
- Distance calculation between vendor and project
- Map display for vendor locations
- Radius filtering visualization
- **Tools:** Google Maps JavaScript API

Acceptance Criteria

- ✓ Admin can import 100+ vendor records via CSV with field validation
 - ✓ Complete CRUD operations for vendors, projects, and tasks
 - ✓ Project documents uploaded and accessible via secure links
 - ✓ Data validation prevents invalid entries (negative budgets, past dates)
 - ✓ Task budget sum validated against project total budget
 - ✓ Google Maps integration calculates accurate distances
 - ✓ Vendor filtering by radius works reliably
 - ✓ All forms have proper error handling and user feedback
-

Phase 4: Bid Invitation System (50 Hours)

Module 4: Automated Vendor Filtering & Invitation

Tasks

Task 4.1: Build intelligent vendor filtering algorithm

- Trade/specialty matching with multiple trade support
- Geographic radius calculation using Google Maps Distance Matrix API
- Insurance validity date checking
- Bonding capacity verification against project value
- Current capacity check (active jobs vs maximum)
- Vendor rating/history filtering
- **Tools:** PostgreSQL queries, FastAPI, Supabase Edge Function, Google Maps API

Task 4.2: Create bid invitation email templates

- Professional HTML email design
- Responsive email layout (mobile-friendly)
- Dynamic content placeholders (vendor name, project details)
- **Tools:** HTML/CSS email templates, React Email

Task 4.3: Implement email service integration

- Configure AWS SES or SendGrid Pro account
- API authentication and credentials
- Email sending function with attachments
- Bounce and complaint handling
- Delivery status webhooks
- **Tools:** AWS SES SDK or SendGrid API

Task 4.4: Custom form setup - See Phase 5

Task 4.5: Build batch email sending function

- Queue-based email processing
- Rate limiting to respect service limits
- Retry logic for failed sends
- Progress tracking and status updates
- Error logging and alerting
- Email personalization engine
- **Tools:** FastAPI, n8n Pro, AWS SES/SendGrid, Redis for queueing

Task 4.6: Implement invitation tracking

- Invitation status tracking
- Submission status updates
- Dashboard status indicators
- **Tools:** Database updates, email webhooks, n8n workflows

Task 4.7: Build vendor selection UI

- Filterable vendor list for task

- Bulk select/deselect vendors
- Preview filtered vendors before sending
- Manual vendor addition/removal
- Send invitation confirmation dialog
- **Tools:** React, UI components

Acceptance Criteria

- ✓ System accurately filters vendors by all criteria (trade, distance, insurance, bonding, capacity)
 - ✓ Google Maps API calculates distances within 50-mile radius accurately
 - ✓ Professional bid invitation emails render correctly across email clients
 - ✓ Unique secure portal links with magic link authentication generated per vendor
 - ✓ All project documents attached to invitation emails successfully
 - ✓ Batch email sending respects rate limits with proper queuing
 - ✓ Invitation status tracked in real-time on admin dashboard
 - ✓ Email failures logged and retried automatically
 - ✓ Admin can preview vendor list before sending invitations
 - ✓ System handles email bounces and unsubscribes gracefully
-

Phase 5: Bid Collection - Custom Secure Forms (60 Hours)

Module 5: Custom Bid Portal with Magic Link Authentication

Tasks

Task 5.1: Design custom bid submission UI

- Multi-step form design (company info, pricing, documents, review)
- Progress indicator component
- Responsive mobile layout
- Form field validation UX
- File upload interface
- Draft save functionality
- Enhanced accessibility features
- Polish and user experience refinements
- **Tools:** React, TailwindCSS, UI/UX design

Task 5.2: Build magic link authentication system

Detailed Breakdown:

- Token generation and encryption
 - Secure random token generation
 - Token hashing and encryption
 - Token expiration logic (24-48 hours)
- Email delivery with magic link
 - Email template with secure link
 - Link formatting and deep linking
 - Resend link functionality
- Token validation middleware
 - Token verification on every request
 - Expiration checking
 - Vendor identity validation
 - Rate limiting on token requests
- Session management for vendors
 - Temporary session creation
 - Session expiration after submission
 - Single-use token enforcement
- Security hardening
 - IP address validation
 - Rate limiting per vendor
 - Suspicious activity detection

Tools: FastAPI, cryptography libraries, JWT, Redis for sessions

Task 5.3: Implement form backend API

- Create bid submission endpoints
- Multi-part form data handling
- Real-time validation during entry
- Draft saving endpoint
- File upload handling
- Final submission endpoint
- **Tools:** FastAPI, Supabase, file handling

Task 5.4: Build form validation logic

- Client-side validation (React Hook Form)
- Server-side validation (Pydantic models)
- Pricing format validation (lump sum, unit pricing)
- Required field enforcement
- Business logic validation (dates, amounts)
- Custom validation rules per field
- **Tools:** React Hook Form, Pydantic, FastAPI

Task 5.5: Implement file upload for vendors

- Drag-and-drop file upload
- Multiple file support
- File type restrictions (PDF, JPEG, PNG)
- File size limits and validation
- Upload progress indicators
- Virus scanning integration (ClamAV)
- **Tools:** React Dropzone, Supabase Storage, file validation

Task 5.6: Build draft save functionality

- Auto-save draft every 2 minutes
- Manual save button
- Draft retrieval on page reload
- Draft expiration after deadline
- Progress preservation
- **Tools:** React, FastAPI, Supabase

Task 5.7: Create submission confirmation flow

- Review page showing all entered data
- Edit buttons to go back to sections
- Final submission button with confirmation
- Submission success page
- Confirmation email generation
- PDF receipt generation
- **Tools:** React, email templates, PDF generation library

Task 5.8: Implement vendor portal routing

- Public routes for magic link landing
- Form access validation
- Redirect logic for expired tokens
- Submission status page
- Error pages (404, 403, expired)
- **Tools:** React Router

Task 5.9: Add vendor-facing help and guidance

- Inline help text for complex fields
- Tooltips and examples
- FAQ section
- Contact support button
- Field-specific validation messages
- Video tutorials or guided walkthroughs
- **Tools:** React, UI components

Acceptance Criteria

- ✓ Magic links work reliably (99%+ success rate)
 - ✓ Tokens expire correctly after 24-48 hours or submission
 - ✓ Vendor cannot access form with expired token
 - ✓ Multi-step form saves progress automatically
 - ✓ Real-time validation provides instant feedback
 - ✓ Vendors can upload insurance certificates and supporting documents
 - ✓ Draft functionality allows vendors to return and continue
 - ✓ Mobile experience is smooth on phones and tablets
 - ✓ Submission confirmation sent immediately via email
 - ✓ Form accessible only via magic link (no direct URL access)
 - ✓ Security testing validates token cannot be reused or forged
 - ✓ File uploads scanned for viruses before acceptance
 - ✓ Submission PDF receipt generated for vendor records
-

Phase 6: Real-Time Dashboard & Response Tracking (60 Hours)

Module 6: Live Bid Status Tracking

Tasks

Task 6.1: Build real-time submission tracking

- Supabase real-time subscriptions setup
- WebSocket connection management
- Submission count live updates ("3 of 7 submitted")
- Status change notifications
- Connection health monitoring
- Fallback to polling if WebSocket fails
- **Tools:** Supabase Realtime, React, WebSocket

Task 6.2: Create bid tracking dashboard

- Overview cards (total invited, submitted, pending)
- Vendor list with submission status indicators
- Timeline view of bid activity
- Submission timestamp display
- Quick actions (resend invitation, view bid)
- Real-time progress bars
- **Tools:** React, Chart.js, UI components

Task 6.3: Implement completeness checker UI

- Visual indicators for complete/incomplete bids
- Missing items list per vendor
- Color-coded status badges
- Completeness percentage
- Action buttons (contact vendor, mark complete)
- **Tools:** React, validation logic

Task 6.4: Build basic charts and visualizations

- Bar chart comparing bid amounts
- Pie chart of submission status distribution
- Timeline chart of submission activity
- Score comparison radar chart
- Interactive tooltips and data exploration
- Export visualizations functionality
- **Tools:** Chart.js, Recharts, data visualization

Acceptance Criteria

- ✓ Dashboard updates within 5 seconds of submission (via WebSocket)
 - ✓ Submission count displays accurately in real-time
 - ✓ Incomplete bids flagged with specific missing items
 - ✓ Charts update automatically without page refresh
 - ✓ Vendor list shows clear status indicators (pending, submitted, incomplete)
 - ✓ Admin can take actions directly from dashboard
 - ✓ WebSocket connection resilient with automatic reconnection
 - ✓ Performance remains smooth with 50+ concurrent bids
 - ✓ No manual refresh needed for any real-time data
-

Phase 7: Automated Reminder System (40 Hours)

Module 7: Deadline Reminders & Escalation

Tasks

Task 7.1: Create reminder scheduling logic in n8n

- n8n workflow for reminder scheduling
- Date calculation logic (T-7, T-3, T-0)
- Timezone handling and conversion
- Vendor status checking (exclude submitted)

- Batch processing for multiple projects
- **Tools:** n8n Pro, date libraries

Task 7.2: Build tiered email templates

- Friendly reminder template (7 days before)
- Urgent reminder template (3 days before)
- Final call template (due date)
- Personalized content per vendor
- Countdown timer in emails
- **Tools:** HTML email templates, dynamic content

Task 7.3: Implement reminder sending workflow

- n8n workflow execution
- Email batch sending
- Status tracking (reminder sent timestamps)
- Error handling and logging
- Delivery confirmation webhooks
- **Tools:** n8n Pro, AWS SES/SendGrid

Task 7.4: Create admin escalation alerts

- Daily summary email to admins
- List of non-responding vendors
- Overdue submission alerts
- Recommended actions
- Dashboard notification badges
- **Tools:** Email templates, n8n Pro

Task 7.5: Build reminder history log

- Display reminder send history per vendor
- Email open and click tracking
- Admin view of all reminders sent
- Resend reminder manually option
- **Tools:** React, database queries

Acceptance Criteria

- ✓ Reminders automatically sent at T-7, T-3, and T-0 intervals
- ✓ Only non-submitting vendors receive reminders (submitted excluded)
- ✓ Escalating urgency in email tone (friendly → urgent → final)
- ✓ n8n workflows run reliably with monitoring
- ✓ Admin receives escalation email after deadline with missing vendor list
- ✓ Reminder history logged in email_log table for audit

- ✓ System handles timezone differences correctly
 - ✓ Manual resend option available from admin dashboard
 - ✓ Reminder send failures logged and retried
-

Phase 8: Bid Comparison & Scoring (40 Hours)

Module 8: Intelligent Bid Analysis

Tasks

Task 8.1: Build bid normalization engine

- Parse different pricing formats (lump sum, unit pricing, mixed)
- Convert all bids to comparable totals
- Handle missing or incomplete pricing data
- Currency and decimal handling
- Line-item breakdown normalization
- **Tools:** JavaScript/TypeScript, FastAPI, business logic

Task 8.2: Implement weighted scoring algorithm

- Price scoring (40% weight, inverse curve)
- Compliance scoring (20% weight, insurance/docs check)
- Past performance scoring (20% weight, history analysis)
- Capacity scoring (10% weight, availability check)
- Timeline alignment scoring (10% weight, start date match)
- Edge case handling (no history, missing data)
- Configurable weights per project
- **Tools:** Python scoring logic, PostgreSQL, FastAPI

Task 8.3: Create side-by-side comparison UI

- Responsive table with all bids
- Sortable columns (price, score, timeline)
- Color-coded indicators (green/yellow/red)
- Expandable line-item details
- Highlight recommended vendor
- Filter and search functionality
- Enhanced mobile-responsive design
- Print-friendly view
- **Tools:** React, data tables library, UI components

Task 8.4: Build recommendation engine

- Automatic highest-scoring vendor identification
- Justification report generation
- Alternative recommendations (2nd, 3rd place)
- Warning flags (over budget, late start, missing docs)
- Override mechanism for admin
- **Tools:** Business logic, report generation

Acceptance Criteria

- ✓ All bids normalized to comparable format regardless of pricing structure
 - ✓ Weighted scores calculated accurately on 0-100 scale
 - ✓ Scoring handles edge cases (new vendors, incomplete data) gracefully
 - ✓ Comparison table displays all bids with complete line-item breakdowns
 - ✓ Highest-scoring vendor recommended with clear reasoning
 - ✓ Admin can sort by any column (price, score, timeline, compliance)
 - ✓ Color-coded warnings show issues (over budget, conflicts, missing docs)
 - ✓ UI is polished, professional, and easy to understand
 - ✓ Export comparison to Excel/PDF for client sharing
 - ✓ Admin can adjust scoring weights per project
-

Phase 9: Award Decision & Contract Generation (40 Hours)

Module 9: Pre-Award Validation & Automation

Tasks

Task 9.1: Build comprehensive pre-award validation

- Start date feasibility check against vendor schedule
- Insurance validity through project completion date
- Bonding capacity validation against project value
- Vendor capacity check (not at maximum jobs)
- Budget variance check (within $\pm 5\%$ of estimate)
- Prior performance validation
- Conflict detection (scheduling, resources)
- **Tools:** Business logic, PostgreSQL, FastAPI, validation engine

Task 9.2: Implement override workflow

- Warning dialog with specific conflict details
- Required justification text field

- Approval routing (if multi-level approval needed)
- Override history logging
- Audit trail for compliance
- **Tools:** React forms, workflow logic, database

Task 9.3: Integrate DocuSign API

- DocuSign account setup and API credentials
- Contract template creation in DocuSign
- Envelope creation and sending
- Signer configuration (vendor, owner)
- Webhook for signature completion
- Document retrieval after signing
- **Tools:** DocuSign API, FastAPI, webhook handling

Task 9.4: Create award letter email template

- Professional congratulations message
- Project details and expectations
- Mobilization instructions and timeline
- Next steps and contact information
- DocuSign link embedded
- **Tools:** HTML email template, dynamic content

Task 9.5: Build contract record creation

- Generate contract record in database
- Store start date, end date, milestones
- Link DocuSign envelope ID
- Update vendor capacity via database trigger
- Create payment schedule (if applicable)
- **Tools:** FastAPI, Supabase, database triggers

Task 9.6: Implement decline notification system

- Professional decline letter template
- Thank you message for participation
- Feedback option for future improvements
- Batch sending to all non-selected vendors
- Maintain positive vendor relationships
- **Tools:** Email automation, batch sending

Acceptance Criteria

- ✓ All validation checks execute automatically before award
- ✓ Validation flags: start date conflicts, expired insurance, insufficient bonding, over capacity,

budget variance

- ✓ Warning dialogs show specific conflicts with details
 - ✓ Admin can override warnings with documented justification
 - ✓ DocuSign envelope created and sent automatically upon award
 - ✓ Award letter email sent with DocuSign link for e-signature
 - ✓ Professional decline letters sent to all non-selected vendors within minutes
 - ✓ Vendor capacity automatically updated via database trigger
 - ✓ Contract record created with complete details (dates, milestones, terms)
 - ✓ Complete audit trail maintained for compliance
 - ✓ DocuSign webhook updates contract status when signed
 - ✓ All database triggers tested and reliable
-

Phase 10: Vendor Commitment & Performance Tracking (50 Hours)

Module 10: Email-Based Milestone Tracking & Automated Alerts

Overview:

Simple email-driven milestone tracking that doesn't overwhelm vendors. No complex portal - just automated check-ins via email with yes/no links, and alerts to PM when intervention is needed.

Tasks

Task 10.1: Build milestone management interface (12 hours)

Admin interface for creating and managing milestones:

- Add milestone to awarded tasks (name, start date, end date)
- Milestone list view with color-coded status on project/task detail page
- Edit and delete milestone functionality
- Simple status tracking: scheduled, started, on_track, delayed, completed
- Notes field for PM to add context about delays or issues
- Milestone detail view showing response history
- **Tools:** React, Supabase API

Task 10.2: Build database schema for milestones (6 hours)

Design and implement milestone tracking tables:

- Milestones table with task linkage, dates, and status tracking
- Milestone responses table logging vendor yes/no answers
- Milestone alerts table tracking all email communications
- Vendor flags table for PM to mark problematic vendors
- Database triggers for auto-updating timestamps and status
- Indexes for efficient date-based queries
- **Tools:** PostgreSQL, Supabase

Task 10.3: Build email response link handlers (10 hours)

Backend endpoints for vendor and PM email interactions:

- Vendor response endpoints handling yes/no clicks from emails
- Token validation ensuring security and single-use
- Response recording in database with timestamp
- Automatic status updates based on vendor responses
- PM date update endpoint with simple form interface
- Date change functionality that resets status and restarts cycle
- Simple HTML confirmation pages ("Thank you, response recorded")
- No response escalation logic (trigger PM alert after 2 days)
- **Tools:** FastAPI, token generation, simple HTML response pages

Task 10.4: Build n8n automated alert workflows (15 hours)

Three core automation workflows:

- **Daily Schedule Checker workflow** (runs 8 AM daily)
 - Check milestones 5 days before start date → send reminder to vendor + PM
 - Check milestones on start date → send start confirmation request to vendor
 - Check milestones 5 days before end date → send progress check to vendor
 - Check milestones on end date → send completion confirmation to vendor
 - Check milestones with no response for 2+ days → escalate to PM
 - Log all alerts sent in database
- **Vendor Response Handler workflow** (triggered by webhook)
 - Process "No" responses from vendors
 - Update milestone status to "delayed"
 - Send immediate alert to PM with vendor contact details and update link
 - Log response in database
- **Completion Notification workflow** (triggered by webhook)
 - Process completion confirmations from vendors
 - Update milestone status to "completed"
 - Send notification to PM that work is finished
 - Log completion in database
- **Tools:** n8n Pro, webhook triggers, database updates

Task 10.5: Create email templates (5 hours)

Seven professional email templates:

- Starting Soon email (to vendor + PM) - 5 days before start
- Start Confirmation email (to vendor) - on start date with yes/no links
- Progress Check email (to vendor) - 5 days before end with yes/no links
- Completion Confirmation email (to vendor) - on end date with yes/no links
- Delay Alert email (to PM) - when vendor reports delay with contact info and update link
- No Response Alert email (to PM) - when vendor doesn't respond for 2+ days
- Completion Notification email (to PM) - when vendor confirms completion
- All templates branded with Blue Onyx branding
- Dynamic content placeholders for project, task, vendor, and date details
- Mobile-responsive design
- **Tools:** HTML email templates, dynamic content system

Task 10.6: Build PM dashboard widgets (7 hours)

Dashboard components for milestone visibility:

- Milestone Timeline View showing visual timeline with color-coded status
 - Gray: scheduled, Blue: started, Green: on_track, Yellow: delayed, Red: no response, Check: completed
- Active Milestones List with filtering and sorting
 - Quick status overview
 - "Update Date" button for delayed milestones
 - Click to view response history
- Alerts/Notifications Panel showing recent activity
 - Delays, no responses, completions
 - Click to view details
 - Mark as reviewed functionality
- Integration with main dashboard for quick overview
- **Tools:** React, Chart.js for timeline visualization, UI components

Task 10.7: Implement vendor flagging system (5 hours)

Manual PM flagging for problematic vendors:

- Flag button on vendor detail page
- Reason dropdown: Missed deadline, Poor quality, Unresponsive, Other
- Notes field for PM to add context
- Flag display on vendor record with date and reason
- Warning icon in vendor lists for flagged vendors
- Flag warnings during vendor selection for new bids
- Unflag functionality when issues resolved
- Link to specific milestone if flag related to milestone issue

- **Tools:** React, Supabase API
-

Acceptance Criteria

✓ **Admin can create milestones** with start/end dates for awarded tasks

✓ **Automated emails sent at correct times:**

- T-5 days before start → reminder to vendor + PM
- On start date → start confirmation to vendor
- T-5 days before end → progress check to vendor
- On end date → completion check to vendor

✓ **Vendor email responses work reliably:**

- Clickable [Yes] [No] links record response
- Simple confirmation page shows after click
- No login/portal required

✓ **PM gets alerted when needed:**

- Vendor clicks [No] → PM gets email with vendor contact info
- No response for 2 days → PM gets escalation email
- Work completed → PM gets notification email

✓ **PM can update end dates:**

- Email link after delay → simple form to pick new date
- Status resets to **on_track** after update
- Automated checks restart with new dates

✓ **No response escalation works:**

- If vendor doesn't click link within 2 days → PM alerted
- Escalation only triggers once per check-in

✓ **Dashboard shows milestone status:**

- Timeline view with color-coded statuses
- Active milestones list with filters
- Recent alerts panel

✓ **PM can manually flag problematic vendors:**

- Flag button on vendor detail page

- Reason + notes captured
- Flag displays in vendor list and bid invitation flow

 All vendor communications logged:

- Audit trail of all emails sent
 - Response history preserved
 - PM actions recorded (date updates, flags)
-

Notes

Key Simplifications from Original Plan:

- No vendor portal for milestone tracking (email-only approach)
- No photo uploads for MVP (can add in Phase 2)
- No complex 4-stage status workflow (just simple states)
- Email-first design throughout
- Human intervention (PM phone calls) handles exceptions

Email Deliverability Critical:

- Must configure SPF/DKIM properly in Phase 12
- Thorough testing of email links required
- Handle bounces and unsubscribes gracefully

Token Security:

- Generate unique secure tokens for each email link
- Tokens expire after 7 days or after first use
- Validate tokens on every request to prevent reuse

Client Philosophy:

- Keep vendors focused on their work, not complex software
 - Automate what can be automated
 - Alert humans only when intervention needed
 - Maintain professional vendor relationships
-

Phase 11: Comprehensive Testing & Quality Assurance (80 Hours)

Module 11: Thorough Testing Across All Layers

Tasks

Task 11.1: Unit Testing

- Vendor filtering algorithm with edge cases
- Bid normalization logic (various pricing formats)
- Scoring algorithm with different scenarios
- Date calculations for reminders and deadlines
- Database trigger testing (capacity, budgets)
- Validation functions (insurance, bonding, dates)
- Magic link token generation and validation
- Milestone check-in date calculations
- Email response link token validation
- Code coverage target: >80% for critical business logic
- **Tools:** Jest, pytest, test fixtures

Task 11.2: Integration Testing

- Custom Vendor Portal → Database full pipeline
- Email delivery pipeline (send → deliver → open tracking)
- Authentication flows (email/password, Google Sign-In, magic links)
- File upload/download to Supabase Storage
- Real-time sync reliability (WebSocket)
- Database trigger chains
- n8n workflow execution (reminder system + milestone alerts)
- API error handling and retry logic
- FastAPI backend with frontend integration
- DocuSign envelope creation and webhook
- Google Maps distance calculation accuracy
- Email response link → database update → PM notification pipeline
- Milestone date update → status reset → automated check restart
- **Tools:** Postman, automated testing, test data

Task 11.3: End-to-End Testing

- Complete bid cycle: create project → add task → invite vendors → receive bids → compare → award → DocuSign
- Complete milestone cycle: create milestone → automated check-ins → vendor response → PM alert → date update → completion

- Multi-user scenarios (concurrent admin actions)
- Mobile responsive testing (real devices)
- Performance testing (dashboard load time, real-time updates)
- Error scenarios (network failures, timeouts, invalid data)
- Magic link access and form submission
- Email link clicks and response recording
- **Tools:** Playwright, Cypress, real device testing

Task 11.4: User Acceptance Testing (UAT)

- Test with 5-7 real test vendors
- Client walkthrough of complete workflows
- Test milestone email responses with real vendors
- Feedback collection and prioritization
- Edge case discovery from real-world usage
- Documentation of issues and feature requests
- Refinement based on client feedback
- **Tools:** Real vendor data, client sessions

Task 11.5: Bug Fixes & Performance Optimization

- Critical bug resolution from testing phases
- Performance optimization (query optimization, caching)
- Error handling improvements
- UI/UX polish based on feedback
- Final regression testing
- Load testing verification
- **Tools:** Debugging, profiling, optimization

Acceptance Criteria

- ✓ All critical functions have passing unit tests (>80% coverage)
- ✓ Integration tests pass for all major workflows
- ✓ Complete bid cycle tested successfully end-to-end with 5-7 test vendors
- ✓ Complete milestone tracking tested with real email responses
- ✓ System tested on mobile (iOS/Android), tablet, and desktop
- ✓ Authentication tested with both email/password and Google Sign-In
- ✓ Magic link authentication tested thoroughly
- ✓ Email response links tested and working reliably
- ✓ Milestone check-in timing tested across multiple scenarios
- ✓ PM date update workflow tested end-to-end
- ✓ FastAPI backend thoroughly tested with all endpoints functional
- ✓ DocuSign integration tested with test envelopes
- ✓ Google Maps integration tested for distance accuracy
- ✓ Known bugs documented, prioritized, and critical issues resolved

- ✓ Edge cases handled gracefully with user-friendly error messages
 - ✓ Performance targets met: dashboard < 2 seconds, real-time updates < 5 seconds
 - ✓ Client confirms system meets requirements through UAT
 - ✓ No critical or high-priority bugs remaining
-

Phase 12: Deployment & Training (60 Hours)

Module 12: Production Launch & User Enablement

Tasks

Task 12.1: Configure production environment

- Deploy backend to AWS (EC2/ECS/Lambda with load balancer)
- Deploy frontend to Vercel Pro
- Configure custom domain and SSL certificates
- Configure production environment variables
- **Tools:** AWS Console, Vercel Pro dashboard, DNS configuration

Task 12.2: Set up production Supabase

- Production database configuration
- Automated backup schedules (daily, weekly)
- Performance monitoring and alerts
- **Tools:** Supabase Pro dashboard

Task 12.3: Configure production email service

- AWS SES or SendGrid Pro production credentials
- Domain verification and authentication
- SPF, DKIM, DMARC records setup
- **Tools:** AWS SES or SendGrid Pro, DNS management

Task 12.4: Set up production n8n workflows

- Production n8n instance configuration
- Workflow deployment and activation
- Credential configuration
- Monitoring and error alerting
- **Tools:** n8n Pro

Task 12.5: Configure monitoring and alerting

- AWS CloudWatch for backend monitoring
- Vercel Analytics for frontend
- Supabase monitoring dashboard
- Error tracking (Sentry or equivalent)
- Uptime monitoring (UptimeRobot)
- Alert notifications (email)
- **Tools:** CloudWatch, Vercel Analytics, Sentry, UptimeRobot

Task 12.6: Create admin documentation

- User manual with annotated screenshots
- Workflow guides for each major process
- Troubleshooting section
- FAQ document
- **Tools:** Documentation software, screenshots

Task 12.7: Create video walkthrough tutorials

- Recording of major workflows (15-20 minutes total)
- Creating a project and inviting vendors
- Comparing bids and making awards
- Managing vendors and documents
- Using the dashboard
- Creating and tracking milestones
- **Tools:** Screen recording, video editing

Task 12.8: Conduct live training session

- Screen sharing demonstration
- Live Q&A with client team
- Hands-on practice session
- Feedback collection
- **Tools:** Zoom, screen sharing

Acceptance Criteria

- ✓ System deployed and accessible via custom domain
- ✓ Frontend on Vercel Pro with optimized performance
- ✓ Backend on AWS with auto-scaling and load balancing
- ✓ Production environment stable and secure
- ✓ Automated daily backups configured and tested
- ✓ Email deliverability verified with SPF/DKIM records
- ✓ Monitoring and alerting active for errors and performance
- ✓ Client can successfully perform all operations independently
- ✓ Documentation covers all major workflows with screenshots
- ✓ Video tutorials available for onboarding new team members

- ✓ Client confirms understanding and satisfaction with system
 - ✓ Post-launch support plan established
-

V. Estimated Hours & Timelines

Phase	Module	Key Tasks	Hours
Phase 1	Foundation & Database	Database schema, RLS, triggers, AWS setup	60
Phase 2	Frontend + Backend + Auth	React, FastAPI on AWS, email + Google auth	100
Phase 3	Entity Management	Vendor/project/task CRUD, Google Maps	60
Phase 4	Bid Invitations	Vendor filtering, email system	50
Phase 5	Bid Collection	Custom portal + magic link authentication	60
Phase 6	Real-Time Dashboard	WebSocket tracking, charts, live updates	60
Phase 7	Reminder System	n8n automated reminders (T-7, T-3, T-0)	40
Phase 8	Comparison & Scoring	Bid normalization, weighted scoring, UI	40
Phase 9	Awards & Contracts	Pre-award validation, DocuSign integration	40

Phase 10	Milestone Tracking	Email-based check-ins, n8n alerts	50
Phase 11	Comprehensive Testing	Unit, integration, E2E, UAT, bug fixes	80
Phase 12	Deployment & Training	AWS/Vercel deployment, docs, training	60
TOTAL			700

Estimated Development Hours: 700 hours (range: 560-840 hours for MVP scope)

Estimated Timeline: 11 weeks (range: 9-13 weeks for MVP scope)

Hours Saved from Original Plan: 110 hours (by simplifying Phase 10)

VI. Estimated Costing

1. Infrastructure Costing (Monthly)

Service	Plan	Cost/Month	Notes
Supabase	Pro	\$25	Database, auth, storage, 100GB bandwidth
Vercel	Pro	\$20	Frontend hosting, serverless functions, analytics

AWS (Backend)	EC2/ECS	\$100 (estimated)	t3.medium or Fargate, load balancer, varies by traffic
AWS SES	Pay-as-you-go	\$10	\$0.10 per 1,000 emails, can use SendGrid instead
SendGrid	Pro (alternative)	-	100k emails/month, if not using AWS SES
n8n	Pro	\$59	Automations, workflows, 300 workflows
Google Maps API	Pay-as-you-go	\$40	Distance Matrix, Geocoding, ~10k requests
DocuSign	Standard	\$30	10 envelopes/user/month, e-signatures
Domain	Annual	\$12	Custom domain (.com)
SSL Certificate	Included	\$0	Provided by Vercel/AWS
Monitoring	Sentry (Team)	\$29	Error tracking, performance monitoring
Backups	AWS S3	\$15	Automated backup storage
Total (Per-Month)		\$340/mo (+/- 20%)	For Complete Deployed System

VII. Additional Notes

Assumptions

- Client provides a complete vendor list with all required data fields (name, trade, location, contact, insurance dates, bonding capacity)
- Project documents are in standard formats (PDF, JPEG, PNG, DWG)
- AWS account with appropriate IAM permissions for infrastructure setup
- DocuSign account for e-signature integration
- Client team available for requirements clarification, UAT testing, and training sessions
- Initial vendor count: 100-200 vendors across all trades
- Typical project: 5-10 tasks requiring bids, 5-7 vendors per task
- Testing includes 5-7 willing test vendors for the UAT phase
- **Milestone tracking uses email-only approach based on client preference**
- **Vendors are not highly technical - simplicity is critical**
- Phase 2 features (Full Scope with advanced analytics, photo uploads for milestones) would be developed after MVP success

Dependencies

- Client weekly meetings to vet the scope or help scoping for the sprint/module
- Timely access to Supabase account credentials and project setup
- AWS account with admin access for infrastructure provisioning
- Email service (AWS SES or SendGrid Pro) account and API keys
- Google Maps API key with Distance Matrix and Geocoding enabled
- DocuSign API credentials and account setup
- n8n Pro account for workflow automations
- Client approval on the database schema design before Phase 1 completion
- Sample project data and documents for testing (provided by Week 2)
- 5-7 test vendors willing to receive bid invitations during UAT
- **Test vendors willing to respond to milestone check-in emails during UAT**
- Custom domain for production deployment
- DNS access for SPF/DKIM email configuration and domain setup
- Vercel Pro and AWS accounts for hosting

Out of Scope (Phase 1 MVP)

The following features are explicitly NOT included in Phase 1 and will be developed in future phases:

- Mobile native apps (iOS/Android)
- Advanced analytics and reporting dashboards
- Automated payment processing and invoicing
- Multi-company/multi-user role management with granular permissions

- Change order management
 - Subcontractor management and tier-2 vendor coordination
 - Budget forecasting and variance analysis
 - SMS notifications via Twilio
 - Weather delay tracking
 - Equipment scheduling and management
 - **Photo uploads for milestone completion (can be added in Phase 2)**
 - **Complex vendor portal for milestone tracking (keeping it email-based)**
-

Success Metrics (90 Days Post-Launch)

Functional Targets

- ✓ 100% of bids collected through standardized format
- ✓ Zero vendor scheduling conflicts post-award due to pre-award validation
- ✓ Real-time dashboard updates with < 5-second data lag
- ✓ 95%+ of project managers use the system for all new bids
- ✓ Complete audit trail for all vendor interactions and decisions
- ✓ DocuSign contracts executed within 24 hours of award
- ✓ **95%+ vendor response rate to milestone check-in emails**
- ✓ **Zero missed milestones due to lack of notification**

Time & Efficiency

- ✓ 70% reduction in time spent on bid management (from 2-3 weeks to 3-5 days)
- ✓ Dashboard consistently loads in < 2 seconds
- ✓ Bid comparison completed in < 5 minutes (vs hours manually)
- ✓ Contract award process in < 10 minutes (vs days manually)
- ✓ Automated reminders eliminate manual follow-up calls
- ✓ **Milestone tracking eliminates manual status update emails from PM**

Quality & Reliability

- ✓ Zero critical bugs reported in the first 30 days post-launch
- ✓ Email deliverability rate > 98%
- ✓ System uptime > 99.5%
- ✓ Form submission success rate > 95%
- ✓ Magic link authentication success rate > 99%
- ✓ Vendor satisfaction score > 4/5 for bid submission experience
- ✓ **Milestone email links work reliably (>98% click-through success)**
- ✓ **PM alerts delivered within 5 minutes of vendor delay notification**

Business Impact

- ✓ Faster project starts due to streamlined vendor selection
 - ✓ Better vendor selection through objective, weighted scoring
 - ✓ Improved vendor relationships through professional communication
 - ✓ Reduced project risk through pre-award conflict detection
 - ✓ Complete compliance documentation for audits
 - ✓ Reduced human cost through these automations
 - ✓ Cost savings in terms of hiring/utilizing human resources
 - ✓ Measurable ROI within 6 months
 - ✓ **Earlier identification of project delays through proactive check-ins**
-

VIII. Conclusion

This **700-hour MVP** delivers a production-ready Bid Management & Vendor Coordination System that will transform Blu OnX Development and Capstone LLC's vendor workflow.

The custom secure form portal approach for bid submission provides a professional vendor experience with complete control, while the simplified email-based milestone tracking respects the technical limitations of construction vendors and avoids overwhelming them with complex portal workflows.

This implementation delivers:

- ✓ Elimination of manual spreadsheet juggling and email chains
- ✓ Secure magic link authentication for vendors
- ✓ Robust FastAPI backend hosted on AWS infrastructure for performance and scalability
- ✓ Professional frontend deployment on Vercel Pro with optimized delivery
- ✓ Intelligent vendor filtering using Google Maps distance calculation
- ✓ Automated bid collection with real-time tracking through a custom portal
- ✓ Objective, data-driven vendor selection through weighted scoring
- ✓ Pre-award validation to ensure vendors can actually perform on required timelines
- ✓ Legally binding contract execution through DocuSign integration
- ✓ **Simple, email-based milestone tracking that vendors actually use**
- ✓ **Automated check-ins that prevent delays without overwhelming vendors**
- ✓ **PM alerts only when human intervention is needed**
- ✓ Complete audit trails for compliance and decision documentation
- ✓ Comprehensive testing ensuring production-ready quality
- ✓ Professional email automation via AWS SES or SendGrid Pro
- ✓ Workflow automation through n8n Pro for reminders and alerts

Key Advantages of This Approach:

For Vendors:

- Simple email-based communication (no complex portals to learn)
- One-click responses to check-ins
- No need to log in repeatedly
- Professional, respectful treatment

For Project Managers:

- Automated tracking without manual follow-up
- Alerts only when intervention needed
- Simple date updates via email links
- Complete visibility into milestone status

For the Business:

- Earlier identification of delays
- Better vendor accountability
- Complete audit trail
- Professional, consistent communication

This approach delivers transformational value and significant time savings for your construction bid management and milestone tracking process with complete control over the vendor experience, while respecting the reality that construction vendors excel at their trade but may not be highly technical.

Total Hours: 700 hours

Timeline: 11 weeks

Monthly Infrastructure Cost: ~\$340

ROI Timeline: 6 months

End of Project Plan